



Fraud Detection in EU Public Contracts

An End-to-End NLP Pipeline for Procurement Risk Assessment

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Legal & Ethical Disclaimer

Academic Purpose Only: This project is an experimental prototype developed solely for academic purposes as part of our Master's degree program. It is not intended for, nor should it be used for, actual legal, financial, or regulatory auditing, compliance, or enforcement.

Experimental AI & False Positives: The models and algorithms utilized in this repository (including NLP similarity scoring and anomaly detection) are experimental. Machine learning models are inherently probabilistic and subject to false positives. A high "risk score" or "anomaly flag" generated by this tool **does not** constitute evidence of fraud, collusion, cartel behaviour, or any illegal activity. It merely indicates a statistical deviation in text structure that warrants human review.

Open Data Usage: The data processed by this tool is sourced from publicly available datasets (OpenTender). The developers of this tool make no claims regarding the accuracy, completeness, or currentness of the underlying public data.

No Accusation of Wrongdoing: Any names of real companies, public authorities, or individuals that appear in the sample data, visualizations, or output of this tool are incidental. This system does not accuse any real-world entity of wrongdoing.

- **Primary Data Source:** Tenders Electronic Daily (TED) — the official EU public procurement journal. It publishes approximately 700,000 notices per year across all 24 EU official languages.
- **Data Access:** Machine-readable subsets are ingested via [OpenTender.eu](https://opentender.eu).



The Problem: Procurement Fraud in the EU

€5–25B

lost annually
to procurement fraud (EU est.)

14%

of EU GDP spent on
public contracts

700k+

contract notices published
across the EU per year

Fraud Typologies Studied (Aura, Constantinescu and Juma, 2021)

Single Bidding

Only one tender received: insufficient competition (Art. 18, Dir. 2014/24/EU)

Winner Concentration

Buyer repeatedly awards to same supplier group (>75% single-bid rate)

Entity Linking

Multiple distinct bidders sharing physical addresses is a strong indicator pre-arranged collusion.

Copy-Paste Description

Near-identical text across different buyers: potential bid coordination signal



Dataset: OpenTender Ireland 2009–2023



148,292

total contract
records

139,629

contracts with
usable text

15 yrs

coverage
2009–2023

Source: OpenTender.eu

Proxy Fraud Labels

Label	Source	Proxy Criterion	Rate
single_bid	OpenTender ind.	INTEGRITY_SINGLE_BID = 0 (0 = red flag, 100 = clean)	~4.2 % of corpus
shared_address_flag**	NLP (spaCy NER)	ORG/LOC entities extracted from text that overlap across distinct contract notices.	~0.17%
winner_concentration	Computed	Buyers in the top 5th percentile of award volume	~5.0%
copy_paste_description	NLP (cos. sim.)	Cosine sim. ≥ 0.92 vs. different buyer	22.6 %

master label target_suspicious = 1 IF (single_bid OR winner_concentration OR copy_paste_description) == 1

01 Preprocessing & EDA

- JSON ingestion & flattening (pandas)
- HTML cleanup, CPV normalisation (regex)
- Fraud label computation from OpenTender indicators
- Token profiling (spacy) · Tokenisation (spaCy sm)

pandas · regex · spaCy

02 Feature Engineering

- TF-IDF matrix (50k features, 1-2 grams)
- Sentence embeddings 768-dim (mpnet-base-v2)
- Cross-buyer cosine similarity scan (≥ 0.92)
- Structured indicators scaling (StandardScaler)

scikit-learn · sentence-transformers

03 Detection & Explainability

- Supervised: Logistic Regression (3 variants)
- Unsupervised: Isolation Forest on embeddings
- RAG audit reports (TF-IDF retriever + Pydantic)
- Error analysis: false positives/negatives

scikit-learn · pydantic



Phase 1: Preprocessing & EDA

1

Language Distribution

98.9% English confirmed in a 5,000-contract sample (manual check).
Language filtering not part of the main preprocessing pipeline

2

Text Cleaning

Regex pipelines strip HTML, reference codes, and boilerplate duplication.

3

Fraud Label Computation

OpenTender indicators mapped to binary proxy labels:
INTEGRITY_SINGLE_BID ≥ 50 $\rightarrow$ clean, $= 0 \rightarrow$ flagged, partial scores $\rightarrow$ excluded

4

Structured Features

buyer_contracts_count, buyer_total_value,
score_integrity/transparency/tender, ind_single_bid,
ind_advertisement_period, ind_procedure_type, ind_call_for_tender

5

EDA

Peak year 2021 ($\sim 10k$ contracts). Top CPV: medical/lab ($\sim 40k$, 29 %),
construction-45 ($\sim 20k$), professional services-71 ($\sim 12k$). Integrity score
bimodal: ~ 67 (clean) and ~ 25 (risk cluster)

Key Outputs

148,292

contracts loaded from JSON

39,022

suspicious (27.95% % of corpus)

98.9 %

contracts in English

~ 10 k

peak contracts in 2021 (max year)



Phase 2: Text Features & Copy-Paste Detection

TF-IDF Baseline

50,000 features · 1–2 grams · sublinear_tf=True

- Shape: $139,629 \times 50,000$ (sparsity 99.82 %)
- Fast, sparse, highly interpretable
- Effective for keyword-overlap detection
- Input to supervised Logistic Regression models

Sentence Embeddings (768-dim)

paraphrase-multilingual-mpnet-base-v2

- Dense semantic representations — captures meaning beyond keywords
- L2-normalised for efficient cosine similarity
- Multilingual: covers non-English notices
- Input to Isolation Forest & LR-Emb models

Copy-Paste Detection Results

≥ 0.92

Cosine similarity threshold

59,374

Flagged cross-buyer pairs

31,584

Unique contracts flagged

22.6 %

Share of corpus flagged

Phase 3: Detection Models

All supervised models use `class_weight='balanced'` to correct for the 29.1 % positive rate and proxy label noise.

Supervised LR — TF-IDF

Logistic Regression on sparse TF-IDF matrix. Baseline with interpretable coefficients that reveal the most predictive procurement terms.

Fast

Interpretable

50k features

Supervised LR — TF-IDF + Structured

TF-IDF concatenated with 9 scaled structured indicators (buyer counts, integrity scores, indicator flags). Best overall performer.

★ Best

PR-AUC 0.741

Multi-feature

Supervised LR — Embeddings + Structured

Dense 768-dim sentence embeddings combined with structured features. Captures semantic similarity beyond keyword overlap.

Semantic

Dense

768-dim

Unsupervised Isolation Forest

Trained on full embedding matrix with no labels — isolates contracts whose language profile is anomalous relative to the corpus.

Label-free

Anomaly det.

10 % cont.



Results & Metrics

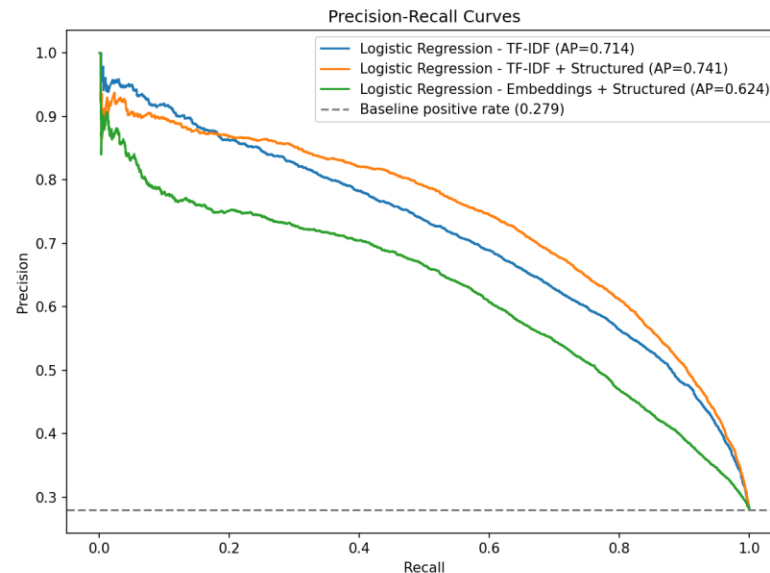
Primary metric: PR-AUC (precision-recall area under curve) — preferred over accuracy given 29.1 % positive rate and label noise from proxy indicators.

Model Comparison

Model	Precision	Recall	F1	PR-AUC	ROC-AUC
LR — TF-IDF + Structured ★	0.637	0.764	0.694	0.741	0.878
LR — TF-IDF only	0.594	0.755	0.665	0.714	0.860
LR — Embeddings + Structured	0.543	0.706	0.614	0.624	0.806
Isolation Forest (unsup.)	0.291	0.104	0.154	—	—

★ Best model · — metric not computed for unsupervised model

PR-AUC Precision-Recall Curves



★ Best Model: LR — TF-IDF + Structured Features

PR-AUC 0.741 · ROC-AUC 0.894 · F1 0.694 · Precision 0.637 · Recall 0.764

Combining TF-IDF text features with 9 structured procurement indicators delivers the strongest risk discrimination.

Phase 4: RAG Explainability Layer

Audit Report Generation Pipeline

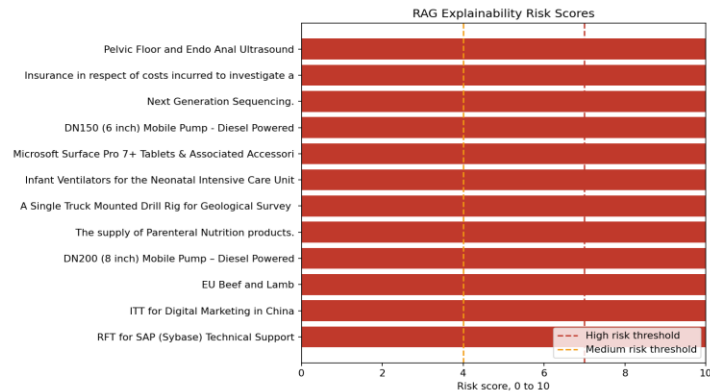
- 1 Flagged contract**
Contract text, buyer, active risk factors
- 2 TF-IDF Retriever**
scikit-learn TF-IDF — retrieves top-3 most relevant knowledge chunks
- 3 Risk Scoring**
Deterministic weighted sum of factor severity (single_bid +3, others +2)
- 4 Pydantic Schema**
Structured JSON output validated against RiskReport schema
- 5 Audit Report**
risk_score, risk_factors, context, explanation, human_review_questions

Sample Report (JSON)

```
{  "title": "Pelvic Floor Ultrasound",  "buyer_name": "HSE",  "risk_score": 10,  "risk_level": "high",  "risk_factors": [    "single_bid", "copy_paste_description",    "embedding_anomaly"  ],  "disclaimer": "Academic prototype..."}
```

Evaluation — 12 Demo Reports

100% schema complete · 100% grounded · 52% context coverage · 3 chunks/report



Current Limitations

Proxy Labels Only

The model is trained on OpenTender indicators and engineered red flags, not verified fraud cases. Outputs are review signals, not fraud classifications.

Short Tender Period Label

The advertisement-period indicator was not useful as a main label because it behaved too uniformly across the Ireland dataset. We keep it visible as context, but exclude it from the main target by default.

No Temporal Split

The evaluation uses a random split. A temporal split, such as training on 2009-2020 and testing on 2021-2023, would better simulate real deployment.

Future Work

-  **Expand from Ireland to more EU countries.**
-  **Improve labels using raw lot-level and bidder-level data.**
-  **Train or fine-tune procurement-specific NER.**
-  **Use a temporal validation split for more realistic evaluation.**
-  **Build dense retrieval over official EU procurement law and guidance.**
-  **Add buyer-supplier graph/network analysis.**



Thank You

Questions & Discussion

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Bibliography

- Aura, A., Constantinescu, A. and Jurma, 'Typologies of EU Fraud', (2021). Available at: <https://eucrim.eu/media/articles/pdf/eucrim-article-2021-029.pdf> [Accessed 5 Dec. 2025].